MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2DSCECC102 – **Fundamentals of OS and Linux**

Duration: 1.5hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall attainability of the outcomes.

Part A

Very Short Answer Questions

Answer **All** Questions

Each question carries **1** mark

1. Write the system software that manages the hardware and software resources of a system. [K][CO1]
2. When a user requests a file from a remote computer, the system retrieves it over the network. What OS component enables this function? [U][CO1]
3. How would you describe the reasons a process shifts from the 'running' state back to the 'ready' state? [U][CO2]
4. What is the name of the scheduler responsible for selecting processes from the ready queue for execution? [K][CO2]
5. Can you identify the scheduling algorithm that runs processes in the order of their arrival? [K] [CO2]
6. How does the Memory Management Unit (MMU) contribute to contiguous memory allocation? [U][CO3]
7. Name the type of fragmentation that occurs due to fixed partition sizes. [K][CO3]
8. What term is used to describe a structured collection of related information stored on secondary storage? [U][CO3]
9. Write the type of access method commonly used in magnetic tape storage. [K][CO3]
10. How would you describe the issue that arises when a file system using contiguous allocation lacks large free spaces? [U][CO3]

[10 x 1 = 10]

Part **B**

Short Essay Type Questions

Answer any **4** Questions

Each question carries **5** marks

11. Summarize the functions of the OS for a user. [U][CO1]
12. List and describe the four necessary conditions for deadlock to occur. [K][CO2]
13. Identify preemptive CPU scheduling algorithms and describe them. [U][CO2]
14. List the common file types recognized by an operating system. [K][CO3]
15. Compare LRU and Optimal page replacement algorithm with examples. [U][CO3]
16. Explain why linked allocation is better suited for files that frequently grow in size. [U][CO3]

[4x5 = 20]

Part **C**

Essay Type Questions

Answer any **2** Questions

Each question carries **10** marks

17. Identify the different types of operating systems. [U][CO1]
18. Discuss a condition that causes a deadlock. Briefly describe the deadlock characterization. [U][CO2]
19. Explain the importance of file operations in file management. List five common types of files in an OS. [U][CO3]

[2 x 10= 20]